Lab instructions

Goal

Introduce an example of a database schema to demonstrate how data can be organized and related in tables.

❖ Task 1

Scenario

This Lab uses the well-known "**chinook sample**" database, which is widely used for example relational database demos and testing purposes. However, you are going to focus only on some of the main tables introduced in this database to highlight how a database schema is designed, and not the entire chinook schema.

Database schema design

Before you develop your actual database, you should design a relevant database schema to document the requirements and to propose an architecture of the database structure. To design a basic database schema, you need to apply the following steps:

Step 1: Define the database purpose.

The "**chinook sample**" database represents a fictitious digital media company that includes information about artists, albums, media tracks, invoices and customers.

Step 2: Identify 6 main tables with a brief description and a primary key for each table.

The chinook sample database has considered adequate normalization level and created 11 tables to store and relate data to avoid data redundancy. However, this exercise will only focus on 6 main tables, including some relevant attributes for each.

Table name	Description	Diagram
Employees	The employee table stores the data of all employees. The diagram presents 8 attributes with relevant datatypes. Employee ID is the primary key in this table.	1
Customers	The customer table stores customers data. In the diagram, 8 attributes with relevant datatypes are presented. Customer ID is the primary key in this table.	2
Invoices	The invoice table stores data on invoices. In the diagram, 5 attributes with relevant datatypes are presented. Invoice ID is the primary key in this table.	3
Artists	The artist table stores data on artists. Only 2 attributes, the artist ID and artist name, are presented in the diagram alongside their relevant data types. Artists ID is the primary key in this table.	4
Albums	The album table stores data about a list of tracks. In the diagram, 3 attributes with relevant data types are presented. Album ID is the primary key in this table.	5

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Tracks	The tracks table stores the data of songs. In the diagram there are 5 attributes with relevant data types. Tracks ID is the primary key in this table.	6

Diagram 1



Diagram 2



Diagram 3

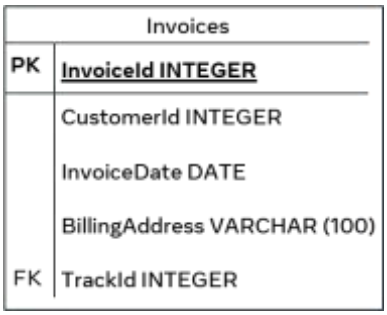


Diagram 4

Artists	
PK	**ArtistId INTEGER**
	Name VARCHAR (120)

Diagram 5

Albums	
PK	**AlbumId INTEGER**
	Title VARCHAR (160)
FK	ArtistId INTEGER

Diagram 6

Tracks	
PK	**TrackId INTEGER**
	Name VARCHAR (200)
	AlbumId INTEGER
	UnitePrice DECIMAL
FK	AlbumId INTEGER

Step 3: Identify the relationships between the 6 main tables.

The chinook sample database defines the following relationships between the 6 stated tables:

- Each employee will support one or many customers.
- Each customer may have multiple invoices
- Each track belongs to one album.
- Each invoice relates to one track.
- Each track belongs to one album
- Each album may contain multiple tracks
- Each artist has one or multiple albums

Instructions

1. Create an entity relationship diagram of the 6 main tables.
2. Write a complete SQL statement to create "**chinook sample**" database and all tables inside the database.
3. Access the devices table that was created inside the "**chinook sample**" database
4. Check the structure of the 6 tables columns and data types.
5. You are required to extend the customized chinook schema by adding a new table called "location" that shows the city and the country the artist lives in.